Applied Soft Computing Journal 150 (2024) 111033

Contents lists available at ScienceDirect

Applied Soft Computing

journal homepage: www.elsevier.com/locate/asoc

Multi-mechanism swarm optimization for multi-UAV task assignment and path planning in transmission line inspection under multi-wind field

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ARTICLE INFO

Keywords:
Transmission line inspection
Multiple UAVs
Task assignment
Path planning
Honey Badger algorithm

ABSTRACT

Along with the growing size of the power system, the inspection and maintenance of power lines and facilities in complicated environments will be put to a greater test. Currently, more and more research are being done on UAV-based transmission line inspection techniques, of which multi-UAV task assignment and path planning are two important aspects. This paper first generates a three-dimensional inspection environment model while considering and measuring the influence of numerous wind fields in the mountain environment. Next, a multi-indicator hybrid cost function that considers wind field influence is defined for the multi-UAV task assignment problem. An improved Bidirectional Ant Colony and discrete Honey Badger Algorithm (BACOHBA) is proposed to solve it. Then, the multi-UAV path planning problem based on the multi-wind field condition is defined, and an improved Honey Badger-Fruit Fly Algorithm (HBAFOA) is proposed to solve it. Finally, simulation experiments are conducted under different algorithms, which prove that the proposed algorithm has better performance in terms of fast-solving ability, solution accuracy, and optimization stability.

1. Introduction

As the power grid covers a wider area, the power lines are getting longer, and the terrain environment they pass through is becoming more and more complex, which puts forward higher requirements for the subsequent power line inspection work. Especially for the transmission lines in the mountainous environment, due to the terrain, climate, and other factors. The traditional manual inspection method is not only inefficient, and in a complex environment may also pose a threat to the personal safety of inspectors. In addition to the geographical environment of the mountain, trees and even weather factors, transmission lines in the towers, and equipment with a certain degree of obscurity, it is difficult to manually find fault points and their fault range. So it is necessary to find a new efficient method to outlaw the old manual inspection method [1].

In recent years, unmanned aerial vehicle (UAV) inspection technology has been paid more and more attention. It has many advantages, like lower costs, convenience, flexibility, safety, reliability, etc. As it can get clear images, and also can reach many areas that the inspector is hard to get, it has a wide range of applications in many fields, such as oilfield inspection [2,3], monitoring the intrusion of obstacles [4,5], disaster relief [6,7], health care [8], agricultural plant protection [9], etc. Many scholars have done a lot of research on UAV inspection of transmission lines and towers, Zhao et al. [10] solved the problem of real-time detection and modeling of power lines and towers respectively and achieved high-precision positioning of lines and towers. Chang et al. [11] explored the UAV trajectory delineation scheme according to different ways of power line inspection, and Wang et al. [12] adopted a collaborative approach of UAV and vehicle to inspect power transmission towers. Esteban et al. [13] discussed the problem of obstacle avoidance when UAVs encountered moving obstacles in flight, and Tong et al. [14] addressed the problem of trajectory planning and online obstacle avoidance of multiple UAVs.

UAV inspection has become the mainstream method of transmission line inspection, and through continuous research and promotion, more methods, theories, and application experiences have been formed [15–17]. At the same time, some problems have also been found. Overhead lines in mountainous environments, often have characteristics such as a large area and a large number of towers, while the rotor UAV, which can perform inspection tasks, has the disadvantage of short endurance [18]. In addition, in the mountain environment, wind is inevitable and its change is large, therefore, the impact of UAV transmission line inspection of wind field factors cannot be ignored. To solve this contradiction, it is necessary to adopt a collaborative inspection method with multiple UAVs. The research on the inspection methods of multiple UAVs can be divided into two parts: task assignment under wind field and path planning under wind field. The task assignment can configure a reasonable number of UAVs for inspection tasks under multiple wind fields, forming the optimal task sequence, while the path

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https://doi.org/10.1016/j.asoc.2023.111033
Received 21 March 2023; Received in revised form 22 September 2023; Accepted 1 November 2023
Available online 10 November 2023
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planning can further find the optimal flight route of the UAV on the basis of the optimal sequence. By solving these two parts with different theoretical methods and connecting them together, a complete multi-UAV inspection method can be formed, which is also the main content of this paper.

For the problem of multiple UAV inspections, a more reasonable way is to allocate multiple task points to multiple UAVs so that each UAV can complete the inspection tasks in an orderly manner. Many scholars have studied the task assignment problem for multiple UAVs: In [19], an immune clone selection algorithm (ICSA) was proposed to solve a multi-base rechargeable multiple UAVs road inspection problem to complete the task assignment to multiple UAVs. In [4], a quantum genetic algorithm was proposed to solve the multi-UAV collaboration problem for global task assignment to multiple UAVs, while the solution was optimized by an optimization strategy using a hybrid frog-hopping algorithm and a simulated annealing algorithm. Jia et al. [20] proposed a multilayer coded genetic algorithm that maintained global search and parallelism capabilities to solve the heterogeneous multi-UAV task assignment problem. Zhang et al. [21] proposed a hybrid algorithm GSA-GA, which combined a gravitational search algorithm with a genetic algorithm to solve the multi-UAV cooperative task assignment problem. These algorithms can effectively solve the problem of reasonably assigning tasks to multiple UAVs in the environment, but there is no detailed treatment of the subsequent flight path of each UAV, i.e., the path planning problem of UAVs.

The UAV path planning problem needs to be considered as an important part of the UAV inspection. Path planning means finding a safe and collision-free flight route between task points and making the UAV fly with the best trajectory, under different required indexes. There are also many effective methods for the UAV path planning problem: Qi et al. [22] solved the UAV path planning problem by introducing dynamic adjustment coefficients so that the UAV would safely and effectively achieve obstacle avoidance. A new spherical vector particle swarm optimization algorithm (SPSO) was proposed in [23] when the UAV was subject to multiple threats in a complex environment. In [24], a multi-strategy fusion differential evolution algorithm (MSFDE) was proposed, which integrated multiple swarm strategies, novel adaptive strategies, and interactive variation strategies to balance the development and exploration capabilities. In [25], an improved whale optimization algorithm (IWOA) was proposed by adding adaptive weights and modifying the linear convergence factor to a nonlinear convergence factor. In [26], based on the pigeon swarm algorithm (PIO) for global optimization, fused with the fruit fly algorithm (FOA) for local optimization, an improved PIOFOA algorithm was proposed to solve the UAV path planning problem. In [27], in order to solve the UAV path planning problem, an improved bat algorithm (IBA) incorporating the artificial bee swarm colony algorithm (ABC) was proposed. In [28], an improved adaptive gray wolf optimization algorithm (AGWO) was proposed for the UAV 3D path planning problem in the earthquake disaster area. In [29], an adaptive particle swarm optimization algorithm (APSO) was proposed to solve the UAV path planning problem, which adaptively adjusted the inertia weights and two learning factors. Li et al. [30] investigated multi-UAV path planning based on multi-objective functions and proposed an improved NSGA-III algorithm. Ge et al. [31] proposed an improved LFOA-NSGA-III algorithm and experimentally verified its effectiveness in the UAV path planning problem. Although most of these studies tend to plan a safe and efficient collision-free path for the UAV between two task points, they are all motivated by the consideration of the UAV's own time consumption, energy consumption, stability, etc. They lack consideration of the influence of external conditions such as multi-wind field problems in the complex mountain environment.

From the above research results, it can be seen that in the UAV task assignment and path planning problem, the swarm intelligence optimization algorithm is getting more and more attention. Swarm intelligence optimization algorithms are derived from the behavior of insects and animals in nature, which has the advantages of being distributed, self-organizing, collaborative, and robust, and are very helpful in solving complex problems [32]. In our study, there are still two problems that need to be solved: (1) In the task assignment part, the optimization performance of the algorithm needs to be further improved, because, under multi-wind field condition, the optimal task allocation result needs better balance ability of the global search and local search of the solving algorithm; (2) most studies only consider influences of multiple UAVs themselves, such as flight time and distance, time consumption, energy consumption, stability, etc., but less consider external influences from the special environment, such as: multiple wind fields in the mountain environment. Honey badger algorithm (HBA) is a new intelligent optimization algorithm proposed by Fatma et al. [33], which simulates the foraging behavior of the honey badger group to find the best. HBA has the advantages of fast convergence speed, high convergence accuracy, and better merit-seeking ability, which can be flexibly combined with different algorithms according to the problem to be solved. Therefore, this paper uses the combination of honey badger algorithm, Ant colony optimization algorithms, and a fruitfly optimization algorithm to solve the task assignment and path planning of multiple UAVs respectively.

The specific contributions of this paper are summarized as follows:

(i) The multi-wind field modeling and regional division of the 3D mountain environment are carried out. On this basis, the modeling and solving of the task assignment and path planning for the multi-UAV transmission line inspection are realized;

(ii) In solving the multi-UAV task assignment problem with multiple wind fields, a discrete honey badger algorithm is proposed and combined with the bidirectional ant colony algorithm;

(iii) The original honey badger algorithm is combined with the fruit fly optimization algorithm to solve the multi-UAV path planning problem with multiple wind fields.

The remaining of this paper is organized as follows: In Section 2, the mountain environment in three dimensions, the transmission line inspection tasks, and the wind field is modeled respectively; In Section 3, the multi-UAV task assignment problem with the multi-wind field is introduced, a BACOHBA algorithm is proposed and its convergence analysis is performed; In Section 4, the multi-UAV path planning problem with the multi-wind field is introduced, an HBAFOA algorithm is proposed; In Section 5, the experiment is designed and many experimental results are got to show comparisons among various algorithms; Section 6 concludes main contributions of our works.

2. Environment modeling

2.1. Mountain terrain modeling

Reasonable establishment of an inspection environment model is the basis of UAV path planning [6]. In this paper, a height matrix is first formed by imitating the data in the elevation map of the real 3D mountain environment. In the environmental spatial modeling problem of unmanned aerial vehicle inspection planning, Digital Elevation Model (DEM) is often used to represent geographic information. This representation method has various advantages such as small data storage and wide information content, which can effectively improve planning efficiency. There are three kinds of DEM grid models, namely, the Contour line model, irregular triangular network model, and regular grid model. The regular grid model is the most widely used model for terrain model building in complex environments. The elevation information in the terrain is represented by a two-dimensional matrix, which is easy to be processed by computers, and has low complexity and computation. This article uses the regular grid model in DEM for environmental modeling.

However, there are fewer height data points in the height matrix, so it needs to be interpolated to get a matrix of higher order $h'(x, y)$, in which a small number of negative points are corrected. Thus the final



Fig. 1. A photo of transmission lines in a real mountain environment.

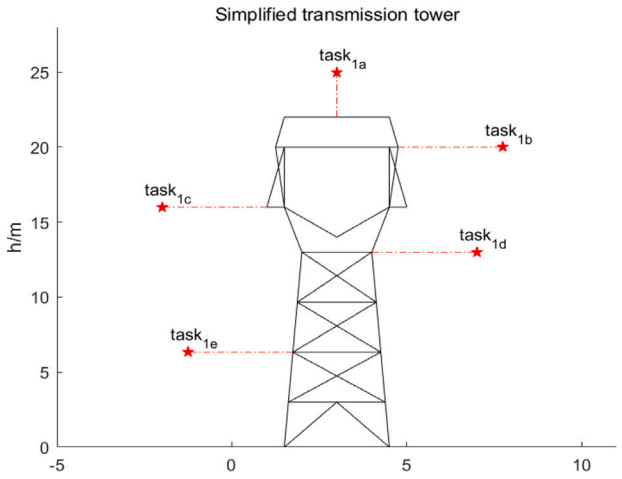


Fig. 2. Simplified transmission tower model.

mountain height matrix is formed as $h(x,y) = max(h'(x,y), h_0(x,y))$, where $h_0(x,y)$ is a zero matrix.

Next, the transmission towers are modeled. Fig. 1 is an overhead view of a set of transmission lines in a real mountain environment [34].

The distribution of transmission towers is often scattered, and the spacing of towers is related to the voltage level, generally ranging from tens to hundreds of meters. The task point to be inspected by the UAV is the scattered transmission towers and long lines in the mountainous environment. Taking the 110 kV level transmission towers as an example, its simplified model is shown in Fig. 2.

These transmission towers are distributed in the mountainous environment model, and their simulation model is shown in Fig. 3.

2.2. Multi-wind field modeling

In a mountainous environment, the impact of wind on the UAV inspection work cannot be ignored. Assuming that the wind direction and wind speed magnitude will not change in a certain area at a certain time, the wind field vector can be expressed as:

$$\vec{V}_w = (V_w, \alpha) \tag{1}$$

where V_w denotes the wind speed and α denotes the wind angle, as shown in Fig. 4.

The wind with angles α of 0°, 90°, 180°, and 270° are called westerly, southerly, easterly, and northerly winds respectively.

In a large mountain environment, different geographical locations often have different wind speeds and directions, thus it will produce different effects on UAVs' task assignment and path planning. At the same time, it will not make a big difference for the wind direction and wind speed in a similar location in the same area. In order to simplify the analysis of the problem, this paper assumes that there are four different wind fields in the mountain environment. Thus four different areas are divided, and the wind field in each area is considered to remain constant over a period of time. The towers to be inspected in each wind field are marked as shown in Fig. 5, in which, the transmission towers in the wind fields Area 1, Area 2, Area 3 and Area 4 are respectively represented by red crosses, yellow asterisks, magenta plus signs and blue circles.

Table 1
Wind field information of different regions.

Area	Wind field (velocity in m/s)
Area 1	$V_w1 = (5.6, 90°)$
Area 2	$V_w2 = (3, 180°)$
Area 3	$V_w3 = (6.2, 0°)$
Area 4	$V_w4 = (5.2, 270°)$

The wind field information of different areas is set in Table 1, and the UAV airspeed is taken as $V_g = 20$ m/s.

The impact of wind on UAVs is mainly reflected in changing their flight speed, affecting their flight time and energy consumption. Firstly, establish the relationship between the wind speed vector and the drone speed vector: The velocity vector of the UAV itself is the airspeed $\vec{V}_a = (V_a, \beta)$, and the real velocity vector obtained by vector summing with the wind speed is the ground speed $\vec{V}_g = (V_g, \gamma)$, and the relationship between them can be represented by the vector triangle in Fig. 6 [35].

In terms of a matrix, this can be expressed as:

$$\begin{bmatrix}\cos\gamma \\ \sin\gamma\end{bmatrix} V_g = \begin{bmatrix}\cos\beta & \cos\alpha \\ \sin\beta & \sin\alpha\end{bmatrix} \begin{bmatrix}V_a \\ V_w\end{bmatrix} \tag{2}$$

Therefore, the formula for calculating the ground speed of UAVs affected by the wind field can be obtained as follows:

$$V_g = V_w \cdot \cos\theta \pm \sqrt{V_a^2 - (V_w \cdot \sin\theta)^2} \tag{3}$$

where $\theta = |\alpha - \gamma|$.

The flight time of a UAV is inversely proportional to its flight speed, and its relationship can be expressed as:

$$time = \frac{D}{V_g} \tag{4}$$

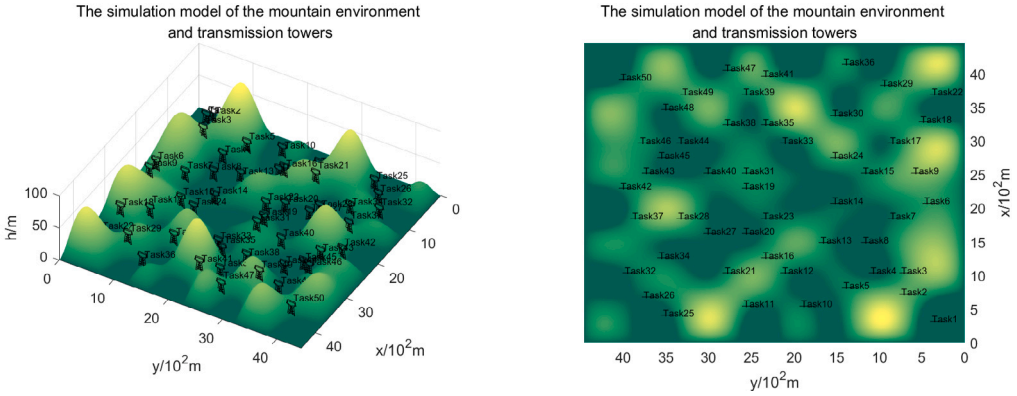
where D denotes the Euclidean distance between two points.

When considering the wind field in the multi-UAV transmission line inspection, it will have a greater impact on the flight speed of the UAV, and then further affects the time cost in the multi-UAV task assignment and path planning. So, in this paper, a reasonable quantification scheme of the impact on the wind field is applied. The reference direction is set based on the south wind 0°, and it is assumed that the coordinates of the two task points or path points before and after are known. So the angle of this flight direction can be regarded as the direction angle of the UAV ground speed γ; judge the wind field area that the task point or path point belongs to, and get the wind speed size V_w and direction angle α; it is assumed that the UAV air speed size V_a is also a known quantity. Among six variables of V_g, γ, V_w, α, V_a and β, the known ones are γ, V_w, α and V_a, and the unknown ones are V_g and β. When calculating the time cost, it only needs to find the ground speed size V_g and divides it with the Euclidean distance between two points. V_g and $time$ can be calculated according to Eqs. (3)–(4).

3. Multi-UAV task assignment considering the effect of wind field

3.1. Problem description

The multi-UAV task assignment problem considering the wind field can be described as follows: n task points are distributed in a map that has been divided into different wind field regions, and it is assumed that the wind field remains constant in each region during this time. m



(a) Distribution map of transmission towers in mountain environment

(b) Top view of transmission towers distribution in mountain environment

Fig. 3. Three-dimensional mountain environment and transmission tower simulation model.

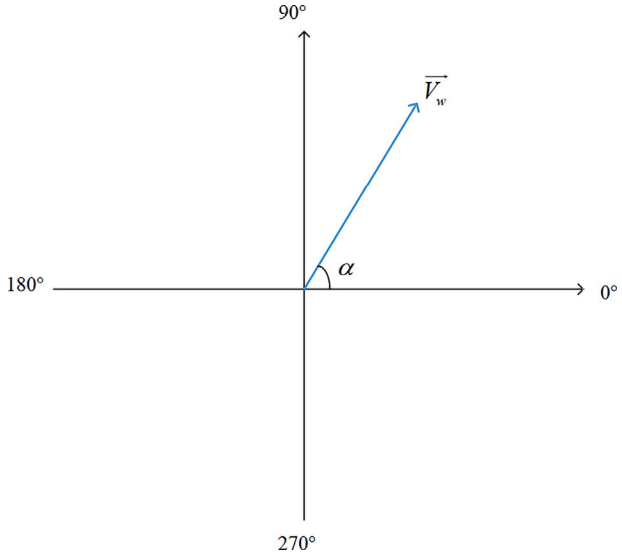


Fig. 4. Representation of the wind field vector.

UAVs visit multiple task points on the map from the same starting point and finally return to the starting point. And each task point needs to be visited once and cannot be repeatedly visited in this process, and the UAVs also have to satisfy their own constraints to obtain the optimal task sequence for each UAV as the basis for the next path planning.

Define the set of transmission towers to be inspected on the map as:

$$Q_N = \{q_1, q_2, \ldots, q_n\} \tag{5}$$

where n is the total number of transmission towers to be inspected, $q_1, q_2, \ldots, q_n$ denote each target tower, each UAV starts from the starting point q_i and finally returns to q_i, $i = 1, 2, \ldots, n$.

Considering this problem a multiple traveling salesman problem (MTSP), a mathematical model is developed for it with the objective function:

$$minF = \sum_{i=1}^{n}\sum_{j=1}^{n} C_{ij}X_{ij} \tag{6}$$

$$s.t.\begin{cases} \sum_{i=q_2}^{q_n} X_{ij} = 1 \quad j \in Q_N \\ \sum_{j=q_2}^{q_n} X_{ij} = 1 \quad i \in Q_N \\ \sum_{i=q_2}^{q_n}\sum_{j=q_2}^{q_n} X_{ij} = n - m - 1 \end{cases} \tag{7}$$

where X_{ij} denotes the set of paths between any two different task points, C_{ij} denotes the flight cost between two task points i and j, and m is the number of UAVs. The constraints indicate that all task points can only be visited once except for the task point which is the starting point $q_i(i = 1, 2, \ldots, n)$; except for the path associated with the starting point q_i, the sum of all paths should be the difference between the number of remaining task points and the number of UAVs, thus each task point is guaranteed to be visited. Take the starting point q_1 as an example, the constraints are:

$$\sum_{i=q_1}^{q_n} X_{ij} = m \quad j = q_1 \tag{8}$$

$$\sum_{j=q_1}^{q_n} X_{ij} = m \quad i = q_1 \tag{9}$$

Eqs. (8)–(9) indicate that all m UAVs will start and end at q_1 and the total number of paths will be:

$$\sum_{i=q_1}^{q_n}\sum_{j=q_1}^{q_n} X_{ij} = n + m - 1 \tag{10}$$

For the flight cost C_{ij}, the cost function between two task points is defined by

$$\begin{cases} F_{Amin}(i,j) = \omega_1 c_1 C_{Adist}(i,j) + \omega_2 c_2 C_{Atime}(i,j) + \\ \qquad\qquad \omega_3 c_3 C_{Achange}(i,j) \\ \omega_1 + \omega_2 + \omega_3 = 1 \end{cases} \tag{11}$$

where ω_1, ω_2, ω_3 are the proportional coefficients of each component cost, c_1, c_2, c_3 are constant.

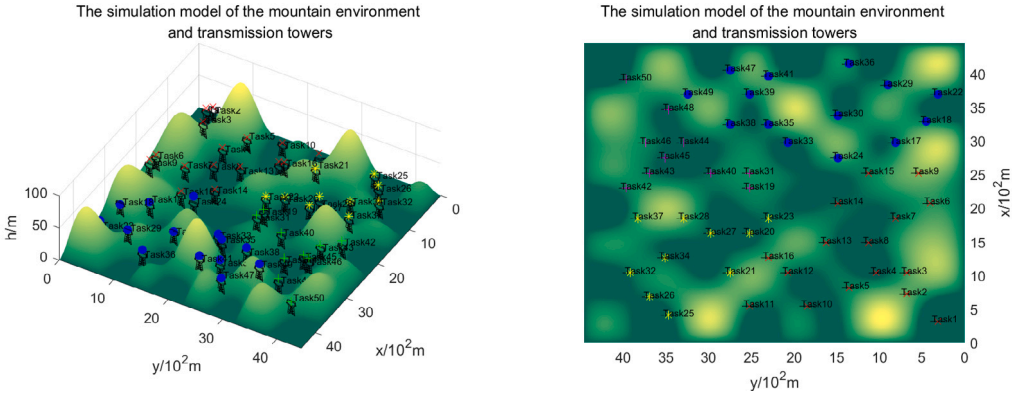
Eq. (11) is used to unify the distance cost, time cost, and wind field variation cost into the same scale. $C_{Adist}(i,j) = \omega_{ij} d_{ij}$ is the distance cost between two points, $d_i j$ is the Euclidean distance between two task points, $\omega_{ij} \in [1, 1.5]$ is the distance weight coefficient between two mission points i and j. When the path between task points does not cross the mountain peaks, $\omega_{ij} = 1$ is set; and when there is a mountain peak obstruction between mission points, for the consideration of the distance cost after UAV obstacle avoidance, the Euclidean distance between two points should be weighted by ω_{ij} based on the height information of the mountain peaks between them.

$C_{Atime}(i,j)$ is the time cost between two points, which is the ratio of the Euclidean distance between two points to the speed of the UAV. It can be calculated by

$$C_{Atime}(i,j) = \frac{d_{ij}}{V_{gij}} \tag{12}$$

$C_{Achange}(i,j)$ indicates whether the wind field changes between two task points, specified as:

$$C_{Achange}(i,j) = \begin{cases} 1 & i \in Area_k, j \in Area_l, k \neq l \\ 0 & else \end{cases} \tag{13}$$



(a) Transmission towers in different wind field areas (b) Top view of transmission towers in different wind field areas

Fig. 5. Distribution map of transmission towers in different wind field areas.

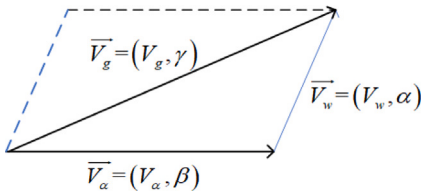


Fig. 6. UAV velocity vector triangle.

where $Area_k$, $Area_l$ denotes the set of task points belonging to different wind field regions.

3.2. Solving multi-UAV task assignment problem by a Bidirectional Ant Colony and discrete Honey Badger Algorithm (BACOHBA)

3.2.1. Honey badger algorithm and its discretization

In recent years, aiming at specific research problems, some novel optimization algorithms have been proposed and applied [36,37]. While in this paper, in order to better solve the multi-UAV task assignment problem, a discretization method [35] is employed to discretize the original continuous honey badger algorithm. In the proposed discrete honey badger algorithm, the honey badger digging mechanism and the honey badger honey harvesting mechanism are both discretized.

(1) Expression of the solution.

In the multi-UAV task assignment problem, different task sequences of multiple UAVs are combined as the location of the honey badger, that is, the solution.

Suppose in a 10 task points and 3 UAVs problem, all UAVs start from task point 1 and are required to return to starting point 1, in which all task points are not repeatedly traversed. One situation is given as:

$$\begin{aligned} Route_1 &= \{1,2,10\} \\ Route_2 &= \{1,3,4,7\} \\ Route_3 &= \{1,5,6,8,9\} \end{aligned} \tag{14}$$

Eq. (14) denotes the task sequence of 3 UAVs, which are combined after removing the common starting point 1 to obtain $X = \{2,10,3,4,7,5,6,8,9\}$. X is defined as the corresponding solution in this situation, and also the location of the honey badger.

(2) Definition of the distance.

The distance between honey badgers is defined as the set of sequential transformation operations, in which the interior points that need to be passed through in the solution, are transformed from one of the positions in the corresponding solution into the other. Supposed there are two honey badgers with the positions of:

$$\begin{aligned} X_1 &= \{2,3,5,10,4,6,7,8,9\} \\ X_2 &= \{2,3,4,5,7,6,10,9,8\} \end{aligned} \tag{15}$$

The distance between two honey badgers is:

$$Dist = \{\langle 3,4\rangle, \langle 5,7\rangle, \langle 7,6\rangle, \langle 10,9\rangle\} \tag{16}$$

where $\langle i,j\rangle$ is called the swap operator, which means that moving the point j at some other positions to closely follow the point i.

(3) Definition of new operation symbols.

The relationship between the position and the distance is denoted as $X_2 = X_1 \oplus Dist$, and the operation symbol $\oplus$ is defined to apply a number of exchange operators of the latter on the former.

The product of the constant coefficient and the distance is denoted by

$$Dist' = A \otimes Dist \tag{17}$$

where the constant A denotes the probability, and the operator symbol $\otimes$ is defined as the exchange operator that takes a part of the latter at random with a certain probability.

(4) Discretization operation.

In the original honey badger algorithm, the position update mechanism is expressed as follows.

In the digging mode, the position of the honey badger is updated by

$$x_{new} = x_{prey} + F \cdot \beta \cdot I \cdot x_{prey} + F \cdot r_1 \cdot \alpha \cdot d_i \cdot |\cos 2\pi r_2 \cdot [1 - \cos 2\pi r_3]| \tag{18}$$

where x_{new} denotes the new location; I denotes the attraction degree; $\alpha = C \cdot exp(-iter/iter_{max})$ denotes the density factor, C is a constant and generally taken as 2, $iter$ and $iter_max$ denote the current and maximum number of iterations; F is a flag that can change the search direction and help jumping out of the local optimum, F is defined by

$$F = \begin{cases} 1 & if\ r_4 \le 0.5 \\ -1 & else \end{cases} \tag{19}$$

where r_1, r_2, r_3 and r_4 are random numbers in the range of $[0,1]$, β represents the ability of honey badger to obtain prey, and its value is generally taken as 6.

In the honey mode, honey badgers are updated by

$$x_{new} = x_{prey} + F \cdot r_5 \cdot \alpha \cdot d_i \tag{20}$$

where r_5 is a random number in $[0,1]$.

A discretization of Eqs. (18)–(20) is got by

$$X_{new} = X_{prey} \oplus [(F \times \beta) \otimes (I \times Dist_0)] \oplus \left\{[(F_1 \times r_1 \times \alpha) \times |\cos 2\pi r_2 \times [1 - \cos 2\pi r_3]|] \otimes Dist_i\right\} \oplus \left\{[(1 - F) \times \alpha \times \beta \times r_4] \otimes Dist_{rand}\right\} \quad (21)$$

$$X_{new} = X_{prey} \oplus [(F \times r_5 \times \alpha) \otimes Dist_i] \oplus \left\{[(1 - F) \times \alpha \times \beta \times r_6] \otimes Dist_{rand}\right\} \quad (22)$$

In the original honey badger algorithm, F denotes a sign to change the search direction, and its value can take 1 or −1. After the discretization of the position update mechanism, the meaning of F remains unchanged and the range of values becomes 1 or 0. Considering $Dist_0$ as the distance between the prey position and the origin, and $I \times Dist_0$ as a whole. In the next combination with the ant colony algorithm, the pheromone concentration of the ant colony algorithm is used to replace the intensity in the honey badger algorithm. This means that all the pheromone concentrations are greater than or equal to the values of the diagonal elements in the pheromone matrix. That is, the front and back points that the ants path through as the selection range of the exchange operator. $Dist_i$ denotes the current distance between the honey badger and the prey, and $Dist_{rand}$ denotes a random set of exchange operators.

3.2.2. Improved BACOHBA algorithm and solution of multi-UAV assignment problem

The honey badger algorithm has the advantages of fast convergence and good searchability. We integrate the position update mechanism of the discrete honey badger algorithm into the ant colony algorithm to improve the solving accuracy. Firstly, the ant colony algorithm is used to obtain the initial solution. As this solution contains the different task sequences of all UAVs, so each ant is considered as m split ant. The corresponding split ants represent the respective task sequences of all UAVs in one case.

It is found that a single direction for search often leads to a high cost of the distance between the last task point and the endpoint in the task point sequence of each split ant; moreover, the value of its own generation varies a lot among different split ants. Hence, in the multi-UAV task assignment problem, it will produce a high and inefficient navigation cost for each UAV when it returns to the uniform endpoint after performing the inspection task, and the cost difference between two UAVs can be large. In this regard, in order to prevent the inefficiency of one-way search, we add a bidirectional search strategy to ant colony optimization algorithms. m positive-order path tables $Table_1$, $Table_2$, …, $Table_m$ and m inverse-order paths $RTable_1$, $RTable_2$, …, $RTable_m$ are generated. Then, combine them and there will be $m!$ combinations; choose the combination with the smallest cost to form the complete path record table $Table_1, Table_2, \ldots, Table_m$. The general idea is shown in Fig. 7, in which different task points are indicated by "#1", "#2", etc.

The path record table of each split ant $Table_1$, $Table_2$, …, $Table_m$ are substituted into Eqs. (21)–(22) to implement the local optimization, and new results $RTable_1$, $RTable_2$, …, $RTable_m$ are got. As the distance between the last task point and the end point of the task sequence generated by the ant colony algorithm is too large, and the self-generation value of different ants is also very different, resulting in poor task assignment results. To solve this, by combining with the update of the discrete HBA, the initial task sequence generated by the ACO with global search ability is further locally searched to improve the solving ability of the algorithm. According to formulas (24)–(25), the individual can be effectively guided to the optimal individual in a "heart-shaped line" by the attraction degree I, and the smooth transition from the digging mode to the honey mode can be ensured by density factor α and honey ability factor β. The solution with the smallest cost is selected as the optimal solution in this generation. Then it is judged whether it is better than the previous generation, and this new solution is brought back to the ant colony algorithm as the basis for an update of the pheromone.

The main steps of solving the multi-UAV task assignment problem by the proposed algorithm are shown below.

Step 1: Establish the environment model, give coordinates of the task points, and set the starting point and end point;

Step 2: Initialize parameters of the Ant Colony Optimize Algorithm (ACO) and the Honey Badger Algorithm (HBA);

Step 3: Search and acquire task sequences by ACO, with m split ants of each ant, containing the sequence of task points for which each UAV should perform inspection tasks;

Step 4: Further optimization of the results obtained from the ACO by the discretized HBA, according to Eqs. (21)–(22);

Step 5: The new optimization results are brought back to ACO for the pheromone updating, and the set of solutions with the smallest cost is selected as the optimal result in this iteration;

Step 6: Repeat Steps 3–5 until the current number of iterations reaches the maximum number of iterations.

The algorithm pseudo code is shown in Algorithm 1.

Algorithm 1 BACOHBA algorithm

Inputs: Task points number n, UAVs number m, cost matrix C, the maximum number of iterations $iter_{max}$, population size $sizepop$, ACO parameters $alpha$, $beta$, rho, Q, HBA parameters β;

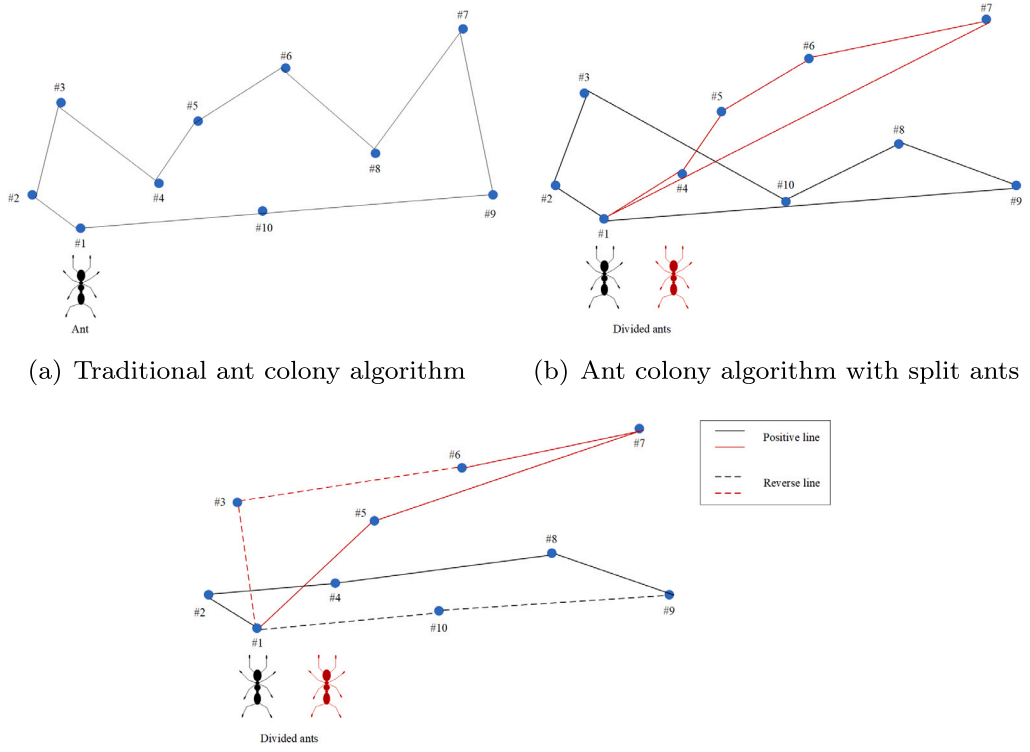
Outputs: Optimized UAVs task sequences, and the minimum cost $Costbest$.

Initialize: The heuristic function matrix Eta, the pheromone matrix Tau, UAVs positive task table $Table_1$, $Table_2$, … , $Table_m$, reverse task table $RTable_1$, $RTable_2$, … $RTable_m$, $iter = 1$;
while $iter \le iter_{max}$ **do**
  Fill the first column of all the rows in the positive task table and reverse task table with 1, which means that the task point 1 as the beginning and end of all UAVs;
  Record numbers of existing task points in the positive task table and reverse task table;
  for $i = 1 : sizepop$ **do**
    while all task points have not yet been assigned **do**
      Find all elements in the first in $Table_k$ and all elements in the first in $RTable_k$, then take the union of these elements as tabu list $tabu$;
      The feasible points set $allow$ is obtained by the complement of $tabu$;
      For m divided ants, select the one with the minimum cost, and put it in the $Table_k$ or $RTable_k$;
    end while
  end for
  Combine positive and reverse tables that minimizes the cost value, update the new results in the positive table;
  Optimize the updated positive table with HBA algorithm, and get new positive table;
  Calculate the total cost value of all UAVs task sequences, and form a matrix $Cost$;
  if $iter == 1$ **then**
    Let $Costbest$ be the minimum in $Cost$, take its corresponding rows in $HTable_1$, $HTable_2$, … $HTable_m$ as $Routebest_1$, $Routebest_2$, … , $Routebest_m$;
  else
    Compare the minimum $Cost$ with $Costbest$, if $Costbest \le \min(Cost)$, remain unchanged, otherwise let $Costbest$ be the minimum in $Cost$;
  end if
  Update Tau;
  Clear the positive task table and reverse task table;
  Let $iter = iter + 1$;
end while

3.2.3. Convergence analysis of BACOHBA

In this section, a Markov chain with no specific constraints [3] is used to prove the convergence of the proposed algorithm.

Definition 1. Consider a discrete sequence of random variables $\{X_t\}_{t \ge 0}$, where t denotes different moments and all random variables X_t $(t = 0, 1, 2, \ldots)$ take the same range of values, called the state space



(a) Traditional ant colony algorithm (b) Ant colony algorithm with split ants

(c) Bidirectional ant colony algorithm with split ants

Fig. 7. The general idea of bidirectional ant colony algorithm.

S. After determining the initial state distribution $P(X_0)$, assume that the conditional distribution between the random state variable X_t at t moment and the random state variable X_{t-1} at the previous moment $t-1$ satisfies:

$$P(x_t|X_0, X_1, \ldots, X_{t-1}) = P(X_t|X_t - 1) \tag{23}$$

That is, X_t only depends on X_{t-1} and not the past random state variables, and this property is called Markovianity, and the sequence of random variables $\{X_t\}_{t\geq 0}$ is called a Markov chain, and the conditional probability distribution $P(X_t|X_{t-1})$ is called the transfer probability distribution of the Markov chain.

Considering more deeply the relationship between the transfer probability distribution of Markov chains and time, we have:

Definition 2. A Markov chain is a homogeneous Markov chain if the transfer probability distribution $P(X_t|X_{t-1}) = P_{X_tX_{t-1}}(t) = P_{X_tX_{t-1}}$ is independent of time.

Theorem 1. *In BACOHBA, the set of solutions at each generation is a discrete homogeneous Markov chain and converges to the global optimal solution with probability 1, and the algorithm converges.*

Proof. First, in BACOHBA, the parameters such as the number of iterations and population size are given, and different iterations are considered as discrete times in Definition 1, and the finite set of constrained solutions is considered as the state space. Meanwhile, according to the individual movement formula, it can be concluded that X_t is a random variable in each generation, and it is only related to the previous state variable X_{t-1}, but not to X_0, X_1, …, X_{t-2} and the number of iterations. So the state of the population is a finite homogeneous Markov chain.

In addition, in BACOHBA, each iteration always retains the best result better than the previous generation, and the evolutionary direction of the population is always in the good direction, satisfying $f(X_t) \leq f(X_{t-1})$.

On this basis, if X_{t-1} belongs to the global optimal solution $k = \{B\}$, then the next state variable X_t also belongs to $k = \{B\}$, which can be expressed by the state transfer probability formula as: $P\{X_t \in k, |X_{t-1} \in k\} = 1$, which is obtained from the full probability formula as:

$$\begin{aligned} P\{X_t \in k\} &= P\{X_{t-1} \in k\} \cdot P\{X_t \in k|X_{t-1} \in k\} + \\ &\quad P\{X_{t-1} \notin k\} \cdot P\{X_t \in k|X_{t-1} \notin k\} \\ &= P\{X_{t-1} \in k\} + (1 - P\{X_{t-1} \in k\}) \cdot P\{X_t \in k|X_{t-1} \notin k\} \end{aligned} \tag{24}$$

For $P\{X_t \in k|X_{t-1} \notin k\}$, assume that the population size is n, that is, there exist n solutions from state X_{t-2} to state X_{t-1} with equal probability of each solution being obtained as $\frac{1}{n}$, and similarly consider that there exist n solutions from state X_{t-1} to state X_t with equal probability of each solution being obtained as $\frac{1}{n}$, then we have:

$$P\{X_t \in k, X_{t-1} \notin k\} = \frac{1}{n} \cdot \frac{n-1}{n} = P\{X_t \in k\} \cdot P\{X_{t-1} \in k\} \tag{25}$$

Let $P\{X_t \in k\} = x$, $P\{X_{t-1} \in k\} = y$, then the equation can be written as $x = y + (1-y) \cdot x$, which simplifies to $y - xy = y \cdot (1-x) = 0$. It is known that $0 \leq y \leq 1$, so it is deduced that $x = 1$, which means that:

$$\lim_{t\to\infty} P\{X_t \in k\} = 1 \tag{26}$$

So, the convergence of the algorithm is proved.

3.3. Performance analysis of the algorithm

Ant colony optimization algorithms have the advantages of strong robustness, strong global search ability, while their own positive feedback mechanism will lead to the disadvantage that it is easy to fall into the local optimal solution. The honey badger algorithm has the advantages of strong local search ability, fast rate of convergence, and so on. The combination of them can avoid the algorithm from dropping in the local optimal solution in the iteration and converge to the global

optimal solution more quickly. However, the complexity of the algorithm will increase with the integration of various mechanisms, which will result in a longer running time compared to other comparative algorithms.

4. Multi-UAV path planning with wind field

4.1. Problem description

After the reasonable task assignment of multiple UAVs, each UAV will get its own task sequence, and will subsequently inspect the task points in order according to this task sequence.

4.1.1. Generation of initial path

A set of initial paths is first generated. For each UAV, in its task sequence two connected task points q_i and q_{i+1} from front to back are taken. The previous task point q_i is used as the starting point of the initial path segment (first path point), from which the UAV flies towards the direction of the target point q_{i+1}. The path point is continuously searched, until the distance between the last path point and the target point is less than a threshold R_{th}, and the UAV can fly directly to the target point [24]. The generation formula of the path point is expressed as:

$$\begin{cases} X_{i+1} = Xi \pm \lambda \cdot \frac{D_x}{R} \cdot (a_1 + a_2 \cdot r_1) \\ Y_{i+1} = Yi \pm \lambda \cdot \frac{D_y}{R} \cdot (a_1 + a_2 \cdot r_2) \\ Z_{i+1} = Zi \pm \lambda \cdot \frac{D_z}{R} \cdot (a_1 + a_2 \cdot r_3) \end{cases} \tag{27}$$

$$\begin{cases} D_x = |X_i - X_{q_{i+1}}| \\ D_y = |Y_i - Y_{q_{i+1}}| \\ D_z = |Z_i - Z_{q_{i+1}}| \\ R = \sqrt{D_x^2 + D_y^2 + D_z^2} \end{cases} \tag{28}$$

where R denotes the Euclidean distance between the current path point and the target point q_{i+1}, λ is the moving step, D_x, D_y and D_z respectively denote the difference of the 3D coordinates between these two points, a_1 and a_2 are two constants controlling the random range of the moving, $a_1 + a_2 = 1$, r_1, r_2 and r_3 are three random numbers, which take values in [0,1].

4.1.2. Cost function

In performing UAV path planning, the flight distance, flight time, flight altitude difference and flight energy consumption are considered as cost indicators [3], as follows.

$$F_{Bmin} = \sum_{i=1}^{m} \sum_{j=1}^{n_i} \sum_{k=1}^{n_{pij}-1} f_B \tag{29}$$

where m is the number of UAVs, n_i is the number of task points assigned to the ith UAV, n_{pij} is the number of path points existing between the jth task point of the ith UAV and the $(j+1)$th task point, and the total cost function is equal to the sum of the costs between the small path segments on the flight path of all UAVs, specifically, the cost between two path points is calculated as:

$$f_B = c_4\omega_4 C_{Bdist} + c_5\omega_5 C_{Btime} + c_6\omega_6 C_{Bpower} + c_7\omega_7 C_{Bheight} \tag{30}$$

where ω_4, ω_5, ω_6, ω_7 are the proportional coefficients of each part of the cost, $\omega_4 + \omega_5 + \omega_6 + \omega_7 = 1$, c_4, c_5, c_6, c_7 are the constant coefficients, the purpose is to unify the distance cost, time cost, energy cost, and height difference change cost to the same scale, C_{Bdist} is the Euclidean distance between two path points, C_{Btime} is the time cost between two path points, $C_{Bheight}$ denotes the height difference between two path points. For the energy consumption cost C_{Bpower}, without considering the influence of wind field, it is often proportional to the flight distance and flight time [24]; but if the wind field is considered, the UAV flight direction and the wind direction need to be judged to determine whether it is flying downwind or upwind. The cost of energy consumption is given as:

$$C_{Bpower} = (c_4\omega_8 C_{Bdist} + c_5\omega_9 C_{Btime}) \cdot (\alpha_1 + \alpha_2 w_{weight}) \tag{31}$$

$$w_{weight} = \begin{cases} \sin\alpha_{gw} & \alpha_{gw} \in [0°, 90°] \\ 2 - \sin\alpha_{gw} & \alpha_{gw} \in [90°, 180°] \end{cases} \tag{32}$$

where α_1, α_2 are constant coefficients, $\alpha_1 + \alpha_1 = 1$, w_{weight} denotes the degree of downwind, calculated from the angle between the UAV flight direction and the direction of the wind α_{gw}, the values range between 0 and 2. When neither downwind nor upwind, that is, when the angle between the UAV and the wind direction is 90°, the energy cost is:

$$C_{BpowerST} = c_4\omega_8 C_{Bdist} + c_5\omega_9 C_{Btime} \tag{33}$$

While the angle is not 90°, the UAV will encounter different degrees of downwind and upwind, and its surrogate value will increase or decrease correspondingly with the angle change, as shown in Fig. 8.

4.2. Solving multi-UAV path planning problem based on honey badger-fruit fly algorithm (HBAFOA)

In the actual use of the honey badger algorithm to solve the path planning problem, the honey badger movement is prone to excessive step size, generating an out-of-bounds situation and forming invalid solutions, which affects the convergence speed and the accuracy of the solution. Therefore, the fruit fly algorithm is considered to be incorporated into the honey badger algorithm, and the solutions that are out of bounds are replanned.

The main steps of HBAFOA for solving the multi-UAV path planning problem are given in below.

Step 1: Establish an environment model, and generate the initial path of each UAV.

Step 2: Set several important parameters of HBA and FOA.

Step 3: Global optimization using the honey badger algorithm: a set of optimal path points is used as the solutions, the initial path point is used as the initial position of the honey badger population, and the position of each honey badger is optimized using the honey badger algorithm to retain its new position after the honey badger movement. The specific calculation formula is as follows.

The attraction degree I_i is:

$$I_i = r_2 \cdot \frac{S}{4\pi d_i^2} \tag{34}$$

$$S = (x_i - x_{i+1})^2 \tag{35}$$

$$d_i = x_{prey} - x_i \tag{36}$$

where S denotes the source intensity, d_i denotes the distance between the prey and the current honey badger individual, and r_2 is a random number in [0,1].

The density factors are:

$$\alpha = C \cdot e^{-\frac{iter}{iter_{max}}} \tag{37}$$

where C is a constant, typically taken as 2, and $iter$ is the number of current iterations.

The location of the honey badger individual is updated by Eqs. (18)–(20).

Step 4: Local optimization using the fruit fly algorithm: the new position is judged to determine whether the path point is moving in a large step or generating an out-of-bounds situation. If not, it remains unmoved. If so, this movement is abandoned; then the position is

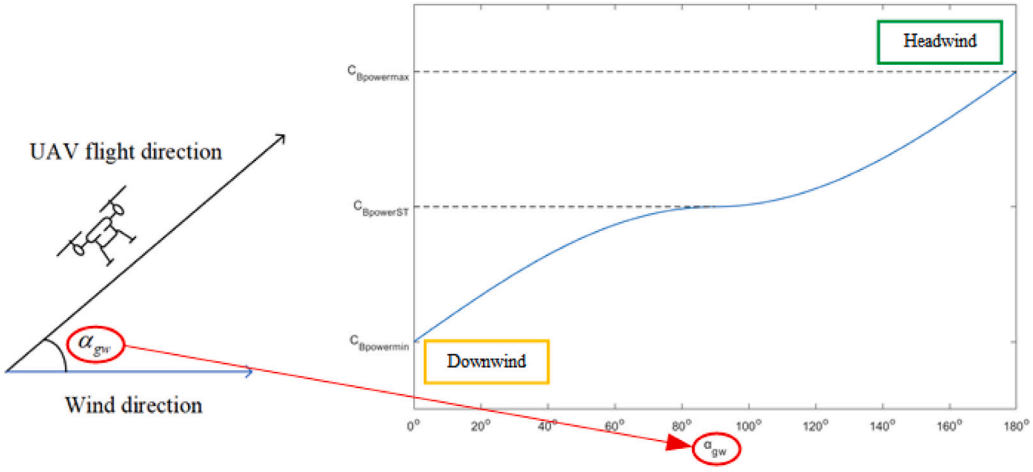


Fig. 8. Relationship between the flight direction of the UAV and the magnitude of energy consumption.

updated and saved by moving from the original position of the honey badger, according to the fruit fly movement formula by

$$\begin{cases} X_{new} = X_{old} + c_1 \cdot \lambda_x \cdot rand() \\ Y_{new} = Y_{old} + c_2 \cdot \lambda_y \cdot rand() \\ Z_{new} = Z_{old} + c_3 \cdot \lambda_z \cdot rand() \end{cases} \tag{38}$$

where $(X_{new}, Y_{new}, Z_{new})$ and $(X_{old}, Y_{old}, Z_{old})$ denote the new and old position of the individual; c_1, c_2, c_3 denote three random numbers in the range [0,1]; $\lambda_x, \lambda_y, \lambda_z$ denote the maximum search step in each direction.

Step 5: In the new location of the honey badger after the local optimization, the cost of each honey badger is solved according to the cost function, and the one with the smallest cost is selected and considered as the optimal honey badger individual for this generation. If the current is the first iteration of the algorithm, the location of this honey badger is recorded as the prey location, that is, the optimal set of path points, and its corresponding generation value is recorded as the minimum cost; if the current is not the first iteration, the minimum cost of this generation needs to be compared with the previous generation, and if the value of this generation is smaller, the result of the previous generation is replaced and recorded, otherwise it remains the same.

Step 6: Repeat Steps 3–5 until the maximum number of iterations is reached.

The pseudo-code of the algorithm is shown in Algorithm 2.

The convergence of the HBAFOA algorithm is proved in the same way as the BACOHBA algorithm in Section 3.2.2.

4.3. Performance analysis of the algorithm

The honey badger algorithm has the advantages of strong search ability, fast rate of convergence, and few adjustable parameters. When it is applied to the path planning problem of multiple UAVs, it is found that the use of its movement formula will often lead to excessive movement of path points or even out of bounds, which will lead to the threat of collision obstacles in the generated path. While the fruit fly algorithm has the advantage of being able to adjust the movement amplitude. In order to address this situation, the fruit fly algorithm is incorporated into the algorithm, filtered out invalid path points, and regenerated them. The new algorithm can greatly improve the effectiveness of generating paths, solve problems quickly, and achieve high accuracy. However, due to the combined mechanism, the new algorithm has the disadvantage of high complexity and long running time compared to other comparative algorithms.

Algorithm 2 HBAFOA algorithm

Inputs: Initial set of path points $Ipathpoint$, cost function F_B, wind field area information $Area_1$, $Area_2$, $Area_3$, $Area_4$, the maximum number of iterations $iter_{max}$, population size $sizepop$, HBA parameters β;
Outputs: Optimized path points $Ppathpoint$, and the minimum cost $Costbest$.

```
Initialize: Initial location of the population, and calculate the initial path cost as the optimal cost Costbest;
while iter ≤ iter_max do
    Calculate the density factor α by Equation (37);
    for i = 1 : sizepop do
        Calculate source strength S, distance d and intensity I by Equations (34)-(36);
        if choose dig next then
            Honey badger individuals move by Equations (18)-(19);
        else
            Honey badger individuals move by Equations (19)-(20);
        end if
        if the newly generated point is out of bounds or too long in stride then
            Discard this move and move it again using the FOA algorithm by Equation (38);
        end if
    end for
    Calculate the cost of all individuals Cost;
    Compare the minimum Cost with Costbest, if Costbest ≤ min(Cost), remain unchanged, otherwise let Costbest be the minimum in Cost;
    iter = iter + 1;
end while
```

5. Simulation experiments

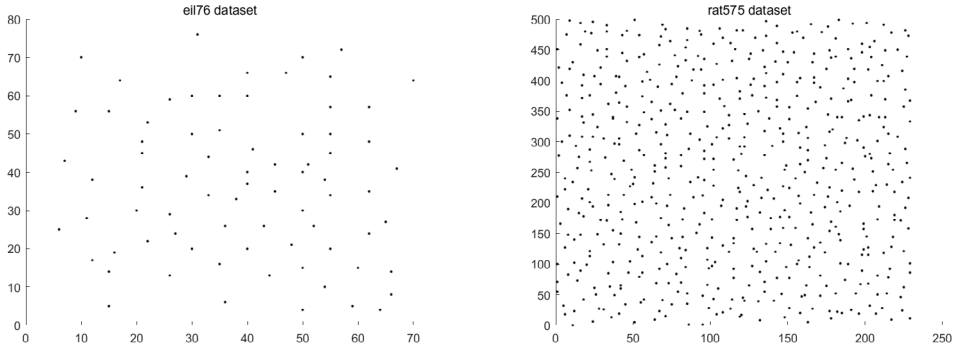
5.1. Case 1: Verification for task assignment method

5.1.1. Experimental environment and data set

In order to prove the effectiveness of the proposed method, an experiment was carried out in MatlabR2016b. The computer configuration is as follows: CPU 2.40 GHz, RAM 4.0 GB. Two public data sets of Traveling Salesman Problem (TSP): eil76 and rat575, are respectively shown in Fig. 9.

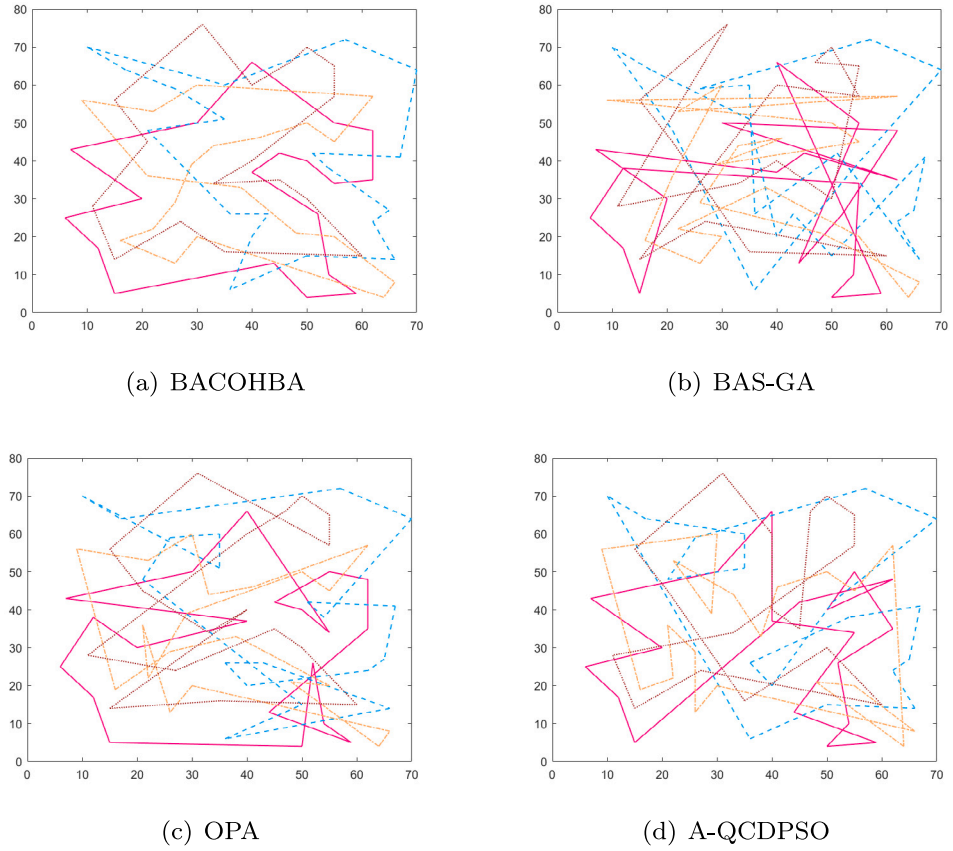
5.1.2. Experiment results and discussion

Different target areas are divided manually to simulate the multi-wind field, and comparative experiments are carried out among A-QCDPSO [38], BAS-GA [12], OPA [25] and the proposed BACOHBA



(a) eil76 (b) rat575

Fig. 9. Public data sets eil76 and rat575.



(a) BACOHBA (b) BAS-GA

(c) OPA (d) A-QCDPSO

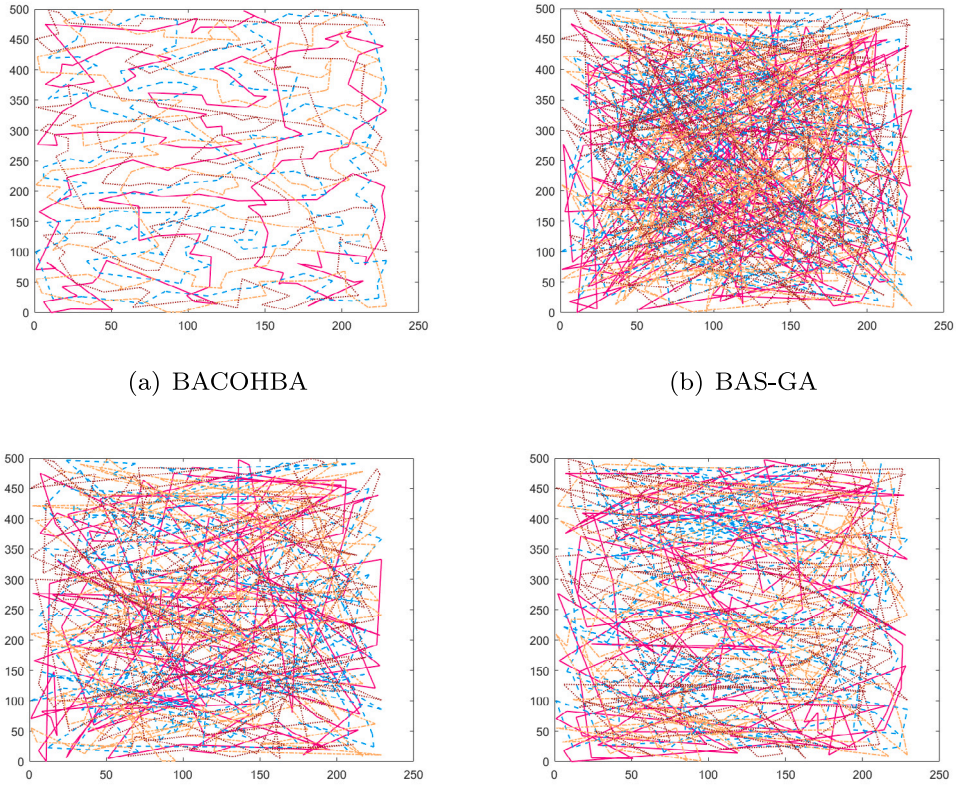
Fig. 10. Simulation results of each algorithm under eil76 data set.

algorithm. Main parameters of different algorithms are given in Table 2, and the optimization results of different algorithms are shown in Figs. 10–11.

It can be seen from Figs. 10 and 11, although the four algorithms (BACOHBA, BAS-GA, OPA, A-QCDPSO) that are compared successfully performed multiple point task assignment (76 points and 575 points), the proposed BACOHBA has the most reasonable task assignment effect, because its task sequences look clear and orderly, and among them, the BAS-GA algorithm has the worst optimization performance.

In order to further prove the effectiveness of the algorithm, more experiment results are presented. Different algorithms are repeatedly run by 20 times, and the evolution curve and performance comparison results are shown in Figs. 12–13 and Table 3.

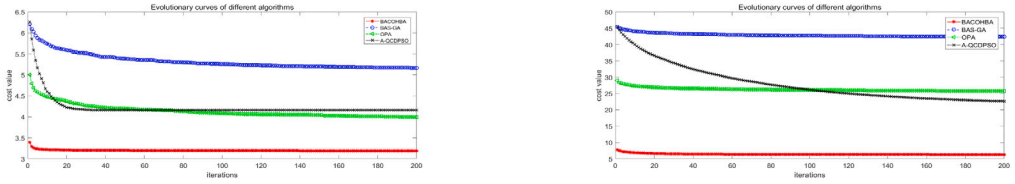
According to the above experiment results, BAS-GA, OPA, and A-QCDPSO algorithms have good results in solving small-scale task assignment problems. However, in the case of large task points, large population sizes and iteration times are often required. Under the same parameter settings, their solution results are shown in Figs. 10 and 11, and it can be seen that they cannot get the relative optimal solution, as shown in Figs. 12–13 and Table 3, These methods have shortcomings in convergence speed and solution accuracy. For the BACOHBA algorithm, the Honey Badger algorithm has a fast convergence speed and strong optimization ability. The discrete Honey Badger algorithm is combined with the ant colony algorithm with a two-way search mechanism. When solving the task assignment problem of multiple UAVs, for the comprehensive cost function with multiple evaluation indicators, it



(a) BACOHBA (b) BAS-GA

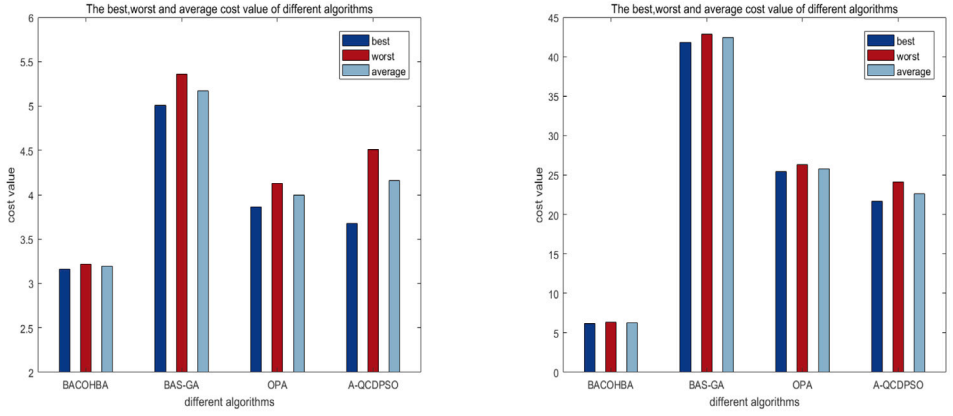
(c) OPA (d) A-QCDPSO

Fig. 11. Simulation results of each algorithm under rat575 data set.



(a) eil76 (b) rat575

Fig. 12. Evolution curves of different algorithms.



(a) eil76 (b) rat575

Fig. 13. Bar chart comparison of simulation results from multiple runs.

Table 2
Main parameter settings and coding methods of different task assignment algorithms.

Task assignment algorithm	Parameter	Coding method
BACOHBA	The population size $sizepop = 20$, the maximum iteration $iter_{max} = 200$, $\alpha = 1$, $\beta = 5$, $\rho = 0.1$, $Q = 1$, the ability of honey badgers to obtain food $\beta = 6$	Designed in Section 3.2.1
BAS-GA	$sizepop = 20$, $iter_{max} = 200$, the mutation probability $Pm_{max} = 9$, $Pm_{min} = 5$	Double chromosomes encoding
OPA	$sizepop = 20$, $iter_{max} = 200$	Nearest-neighbor based multi-traveling salesman problem
A-QCDPSO	$sizepop = 20$, $iter_{max} = 200$, $c_1 = 1.5$, $c_2 = 1.5$, $w = 0.8$	Two-dimensional task allocation matrix and sequence matrix

Table 3
Performance comparison of different algorithms.

Data set	Algorithm	Cost value				Running time (s)
		Best	Worst	Ave	Std	
eil76	BACOHBA	3.1612	3.2145	3.1909	0.0135	17.4
	BAS-GA	5.0112	5.3573	5.1701	0.0953	0.5
	OPA	3.8620	4.1263	3.9947	0.0623	6.2
	A-QCDPSO	3.6733	4.5115	4.1561	0.1942	1.0
rat575	BACOHBA	6.1952	6.3878	6.3022	0.0401	230.0
	BAS-GA	41.7870	42.8132	42.3750	0.2995	1.2
	OPA	25.4466	26.2793	25.7922	0.1843	85.2
	A-QCDPSO	21.6326	24.0866	22.6529	0.7007	1.8

Table 4
The parameter setting for different task assignment algorithms.

Algorithm	Parameter
BACOHBA	The population size $sizepop = 30$, the maximum iteration $iter_{max} = 100$, $\alpha = 1$, $\beta = 5$, $\rho = 0.1$, $Q = 1$, the ability of honey badgers to obtain food $\beta = 6$.
ACO	The population size $sizepop = 30$, the maximum iteration $iter_{max} = 100$, $\alpha = 1$, $\beta = 5$, $\rho = 0.1$, $Q = 1$.

can be seen from Figs. 10 and 11 that the task assignment sequence calculated by the BACOHBA algorithm is significantly superior to the sequence obtained by other algorithms. At the same time, Figs. 12–13 and Table 3 also show that the BACOHBA algorithm is superior to other algorithms in terms of convergence speed, solution accuracy, and stability in 20 repeated experiments. However, in order to get better results, the BACOHBA algorithm sacrifices its complexity, resulting in a relatively long running time.

5.2. Case 2: Verification for path planning method

5.2.1. Experimental environment and parameter setting

To demonstrate the effectiveness of the proposed method, experiments were conducted in MatlabR2016b with the following computer configuration: CPU 3.40 GHz, RAM 8.0 GB. Multiple UAVs started from the same point, the starting position was set at (315,315,25), and the number of task points was set to $n = 50$.

For record purposes, the set of experiments using algorithm A to solve the task assignment problem and algorithm B to solve the path planning problem is referred to as "A+B". Next, the BACOHBA + HBAFOA proposed here will be compared with ACO + AGWO [28], ACO + APSO [29], ACO + HBA [33], ACO + IBA [27], ACO + ORPFOA [3], and ACO + PIOFOA [26]. 30 independent experiments are repeatedly conducted for each group of algorithms, and the parameter setting for different task assignment algorithms is given in Table 4, for different path planning algorithms is given in Table 5.

5.2.2. Experiment results and discussion

The evolutionary curves, generation value comparisons, and optimization performance of the results are shown in Figs. 14–15 and Table 6, which are obtained from different optimization algorithms in

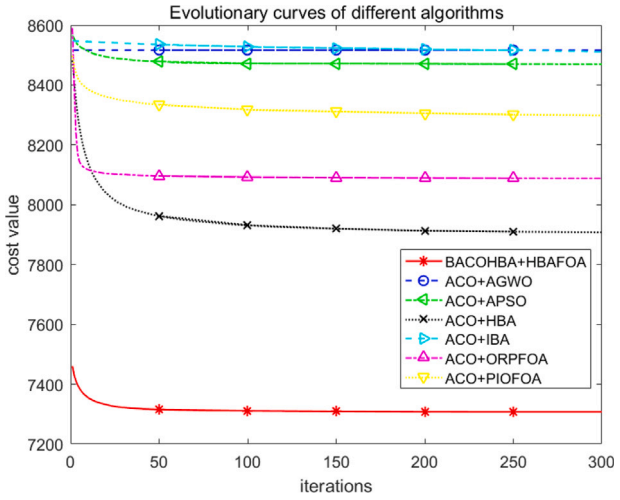


Fig. 14. Evolutionary curves of different algorithms.

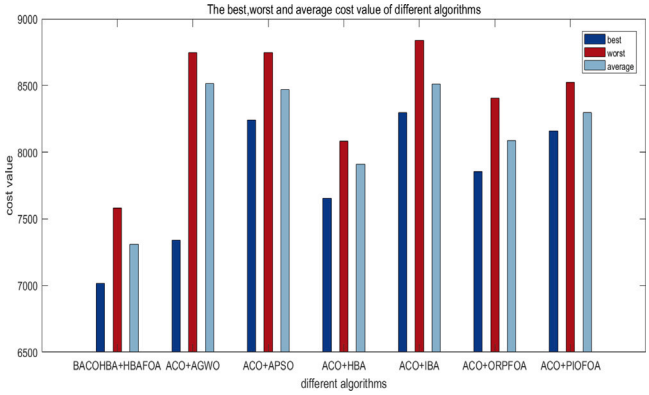


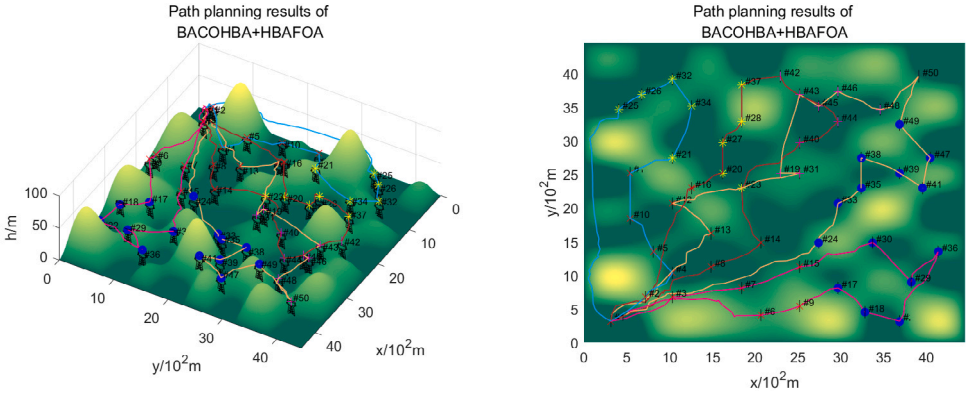
Fig. 15. Comparison of best, worst and average results of different algorithms.

solving the task assignment and path planning problems for multi-UAV inspection of transmission lines.

On the basis of results in Figs. 14–15 and Table 6, more detailed discussions are given in the below, where the transmission towers, that is, the task points, are represented by "#1", "#2", etc.

(1) BACOHBA+HBAFOA

According to Figs. 14–15, it can be seen that the HBAFOA algorithm is significantly better than other algorithms in terms of convergence speed and solution accuracy; and Table 6 can show that in 30 independent experiments, the HBAFOA algorithm performs well in terms of stability performance, and the corresponding Std value is lower; in terms of fast solution capability, the HBAFOA algorithm converges faster, and the running time is also reduced by 51% on the basis of traditional HBA algorithm. However, the algorithm complexity is



(a) Path planning results of BACOHBA+HBAFOA (b) Aerial view of path planning results

Fig. 16. Path planning results in mountain environment using BACOHBA+HBAFOA.

Table 5
Main parameter settings and coding methods for different path planning algorithms.

Algorithm	Parameter	Coding method
HBAFOA	$sizepop = 30$, $iter_{max} = 300$, the ability of honey badgers to obtain food $\beta = 6$.	Each path with a certain number of position points in the three-dimensional space is regarded as a particle, and the flight region is regarded as the solution space of each particle. The process of path planning can be regarded as the process of finding the optimal position of a population containing a certain number of particles in the solution space according to constraints and cost functions.
AGWO	$sizepop = 30$, $iter_{max} = 300$.	
APSO	$sizepop = 30$, $iter_{max} = 300$, self cognitive item learning factor $C_1 \in [1, 3]$, group cognitive item learning factor $c_2 \in [1, 3]$.	
HBA	$sizepop = 30$, $iter_{max} = 300$, the ability of honey badgers to obtain food $\beta = 6$.	
IBA	$sizepop = 30$, $iter_{max} = 300$, frequency $f \in [0, 2]$, acoustic loudness coefficient $A_0 = 0.9$, pulse frequency coefficient $r_0 = 0.5$.	
ORPFOA	$sizepop = 30$, the maximum iteration $iter_{max} = 300$.	
PIOFOA	$sizepop = 30$, $iter_{max} = 300$, the map and compass factor $R = 2$.	

Table 6
Performance comparison of different algorithms.

Algorithm	Cost value				Running time (s)		
	Best	Worst	Ave	Std	Best	Worst	Ave
BACOHBA + HBAFOA	7018.3	7581.5	7307.4	105.18	351	409	378
ACO + AGWO	7662.2	8748.0	8516.1	132.43	50	59	54
ACO + APSO	8242.1	8745.2	8469.8	112.85	50	62	56
ACO + HBA	7652.8	8083.6	7907.9	108.98	707	827	770
ACO + IBA	8294.2	8836.7	8511.1	126.89	52	61	57
ACO + ORPFOA	7855.8	8407.2	8087.8	114.56	81	102	92
ACO + PIOFOA	8160.7	8522.5	8298.3	97.06	142	165	154

sacrificed for better solution accuracy and stability, resulting in a longer running time compared with other algorithms. For the HBAFOA algorithm, the honey badger algorithm itself has a high global search capability and search accuracy, and the FOA algorithm is incorporated on its basis to enhance its local search capability. When solving the specific path planning problem, for the comprehensive evaluation function with multiple evaluation indexes, the convergence speed of the solution problem is faster and the solution accuracy is higher, which can effectively solve the problem, and the results are shown in Fig. 16.

(2) ACO+AGWO

According to Figs. 14–15 and Table 6, the AGWO algorithm performs poorly in terms of solution accuracy and algorithm stability, and is ineffective for solving the specific problem presented in this paper. For the AGWO algorithm, despite the adaptive adjustment of the convergence factor and weight factor based on the gray wolf algorithm GWO, the stability in the optimization in solving the path planning problem is not good, and the results are shown in Fig. 17.

(3) ACO+APSO

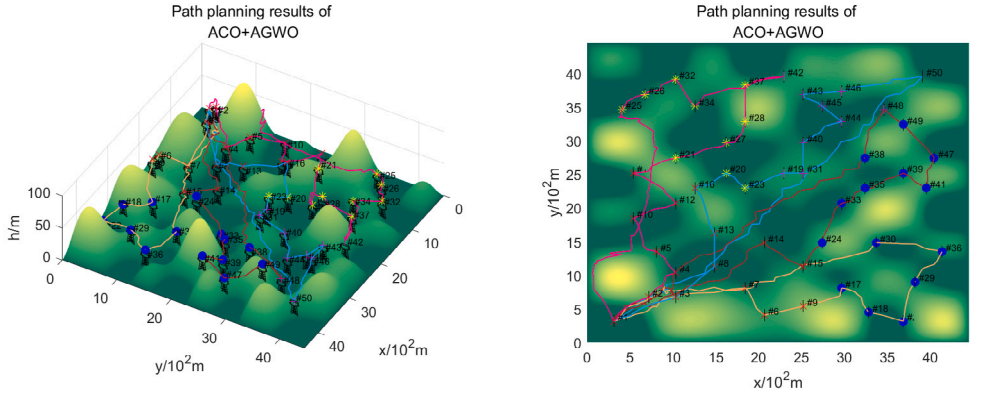
According to Figs. 14–15 and Table 6, the APSO algorithm performs poorly in terms of solution accuracy and algorithm stability, and is ineffective for solving the specific problem presented in this paper. For the APSO algorithm, although the inertia weights and learning factors in the particle swarm algorithm PSO are adaptively adjusted, and the global search capability and search accuracy are improved compared with the classical particle swarm algorithm, it is difficult to obtain the global optimal solution compared with other algorithms, and the results are shown in Fig. 18.

(4) ACO+HBA

According to Figs. 14–15 and Table 6, the convergence speed of HBA algorithm is slow and the running time is long, while the final obtained generation value is significantly smaller and the solution accuracy is better compared with several other algorithms. For the HBA algorithm, the global search capability and search accuracy are relatively good, but when applied to the path planning problem, the convergence speed is slow and there is room for further improvement, and the results are shown in Fig. 19.

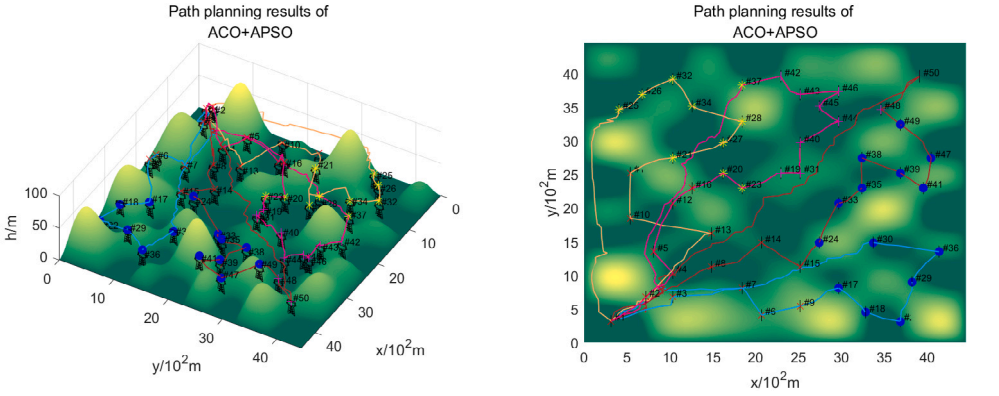
(5) ACO+IBA

According to Figs. 14–15 and Table 6, the IBA algorithm performs poorly in terms of solution accuracy and algorithm stability, and is ineffective for solving the specific problem presented in this paper. For the IBA algorithm, the idea of the ABC is incorporated into the BA of the



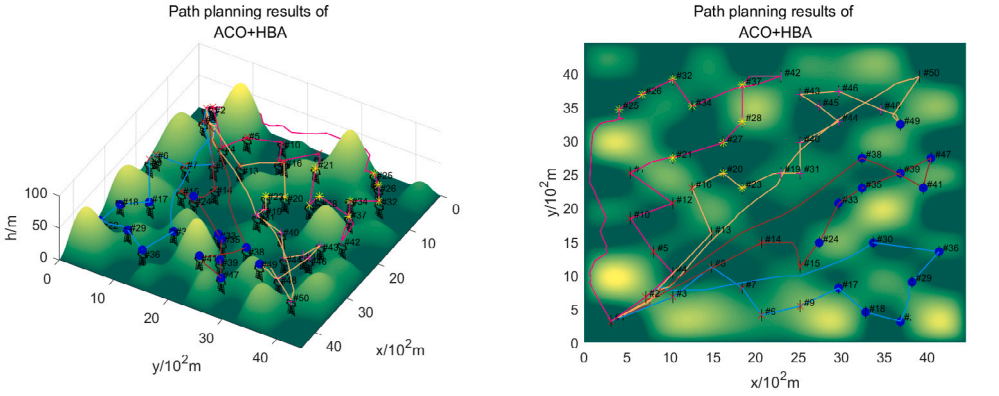
(a) Path planning results of ACO+AGWO (b) Aerial view of path planning results

Fig. 17. Path planning results in mountain environment using ACO+AGWO.



(a) Path planning results of ACO+APSO (b) Aerial view of path planning results

Fig. 18. Path planning results in mountain environment using ACO+APSO.



(a) Path planning results of ACO+HBA (b) Aerial view of path planning results

Fig. 19. Path planning results in mountain environment using ACO+HBA.

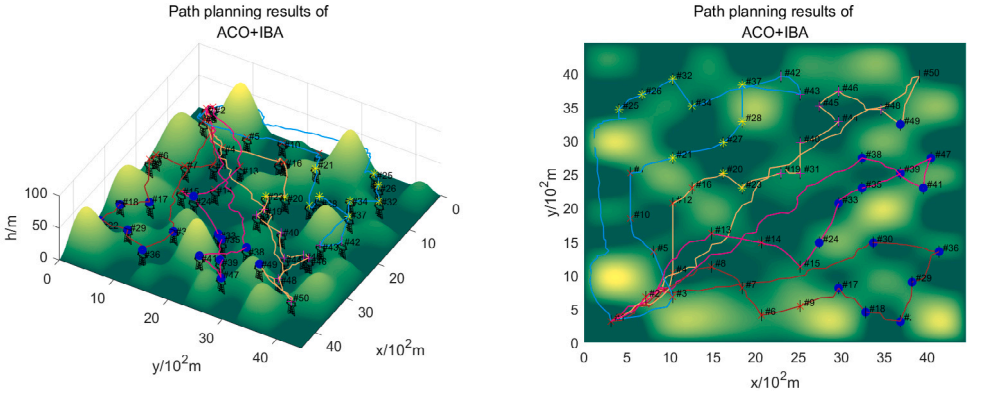
bat algorithm, and despite the improvement in local search capability, the solution accuracy is not high in the path planning problem with multiple evaluation metrics, and the results are shown in Fig. 20.

(6) ACO+ORPFOA

According to Figs. 14–15 and Table 6, the ORPFOA algorithm converges faster and the final obtained generation value is smaller, but the limitation of its search mechanism makes its final convergence of the obtained results often not optimal, which is not effective for solving the specific problem proposed in this paper. For the ORPFOA algorithm, which incorporates the idea of reference points, the convergence speed of the algorithm is relatively fast, but after considering the influence of wind field in the path planning problem, its solution accuracy is not high, and it is difficult to get the global optimal solution, and the results are shown in Fig. 21.

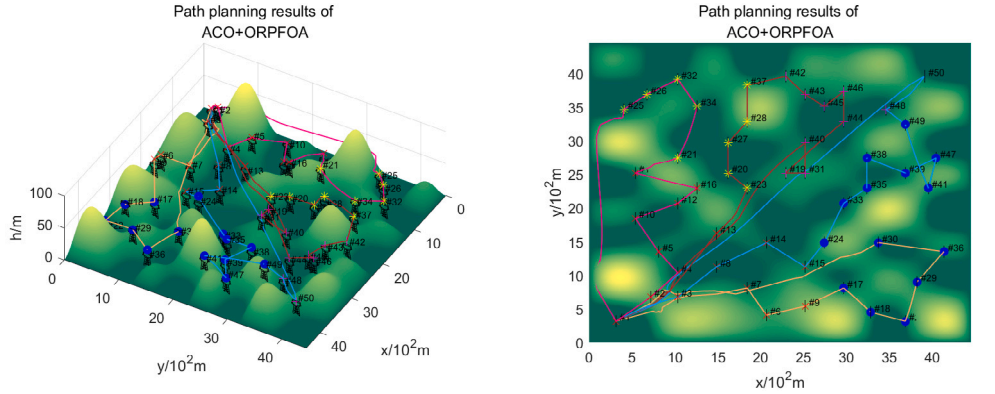
(7) ACO+PIOFOA

According to Figs. 14–15 and Table 6, the PIOFOA algorithm performs poorly in terms of solution accuracy and algorithm stability, and is ineffective for solving the specific problem presented in this paper.



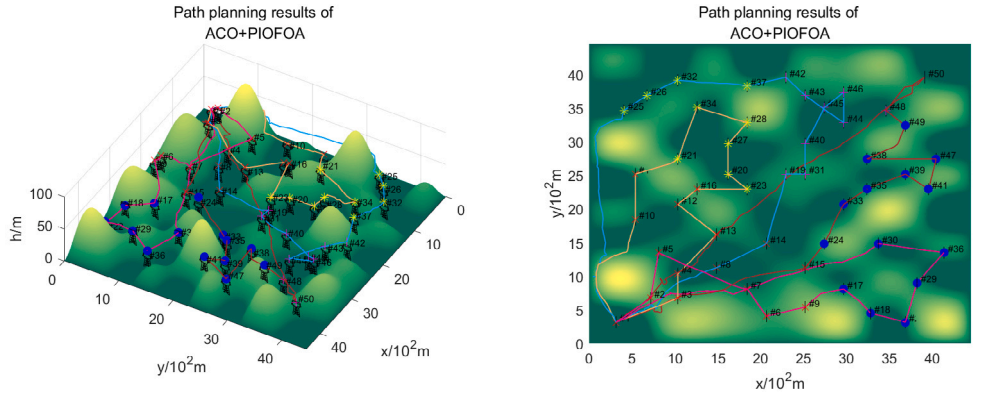
(a) Path planning results of ACO+IBA (b) Aerial view of path planning results

Fig. 20. Path planning results in mountain environment using ACO+IBA.



(a) Path planning results of ACO+ORPFOA (b) Aerial view of path planning results

Fig. 21. Path planning results in mountain environment using ACO+ORPFOA.



(a) Path planning results of ACO+PIOFOA (b) Aerial view of path planning results

Fig. 22. Path planning results in mountain environment using ACO+PIOFOA.

For the PIOFOA algorithm, the pigeon swarm optimization algorithm is incorporated to improve the fruit fly algorithm, which improves the global search ability and local search ability, but like ORPFOA, its optimization effect is not good after considering the influence of wind field in the path planning environment, and the results are shown in Fig. 22.

In summary, the performance comparison of the above algorithms is summarized in three aspects.

- **Solving speed.** In terms of running time, it can be seen from Table 6 that the ACO+HBA algorithm has the longest running time, because its population diversity is richer, and in order to find

the optimal solution with better effect makes the computational complexity higher, while BACOHBA+HBAFOA incorporates the new algorithm and improves the original algorithm on its basis, which speeds up the convergence speed and greatly reduces the running time. Although the running time is still more than the other algorithms, the solution effect is better because of its more comprehensive consideration of the integrated cost function and make overall complexity increased.

- **Precision.** In terms of solution accuracy, it can be seen from Table 6 that among the 30 experiments, the solution obtained by BACOHBA+HBAFOA has the smallest cost at the best, the worst and the average, and its convergence speed is faster and the solution accuracy is higher as shown by the evolution curve in Fig. 18.

- **Stability.** In terms of optimization stability, it can be seen from Table 6 that among the 30 experiments, except for the ACO+PIOFOA algorithm, the average variance value Std of the BACOHBA+HBAFOA algorithm is the smallest, and it is not much different from the worst variance value in the comparison experiments, and the stability of the results is good under the condition that the obtained solution generation value is as small as possible, which can effectively solve multi-UAV task assignment and path planning solution problem.

6. Conclusions

In this paper, a new method is proposed to solve the problem of task assignment and path planning for multiple UAV in a three-dimensional mountainous environment considering the influence of the wind field. In order to verify the effectiveness of the method, a mountainous environment is firstly modeled, the wind field area in the inspection site is divided, the number of UAVs is set, and the quantification scheme of the impact on the wind field is determined. Next, in the task assignment part, the idea of the ant colony algorithm is improved to fuse with the discrete honey badger algorithm to form BACOHBA and apply it to the scene, and the comprehensive evaluation function is set that considers the distance, time, and crossing mountain peaks. Then, using the task sequence of each UAV obtained in the task assignment stage to continue to solve the path planning problem, and the improved HBAFOA algorithm is proposed, and the comprehensive evaluation function formed by considering multiple factors of distance, time, altitude difference change, and energy consumption is set to realize the path planning of multiple UAVs. Finally, the proposed new algorithm is analyzed for convergence and compared with other improved swarm intelligence algorithms proposed in recent years to demonstrate the effectiveness. Our future work will focus on reducing the running time of the algorithm and on the quantification of the wind field in the UAV path planning problem in a more refined way.

CRediT authorship contribution statement

Kun Li: Conceptualization, Methodology, Software, Validation, Writing – original draft, Funding acquisition. **Xinxin Yan:** Conceptualization, Writing – original draft, Software. **Ying Han:** Conceptualization, Methodology, Software, Funding acquisition.

Declaration of competing interest

The authors declared that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Data availability

Data will be made available on request.

Acknowledgment

This work was supported by the LiaoNing Revitalization Talents Program (XLYC2007091); this work was also partly supported by the National Natural Science Foundation of China (Grant Number 62203197).

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